Plant Disease Classification: A Comparative Analysis of Baseline and ANention-Enhanced Deep Learning Models

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Automated plant disease detection is essential to precision agriculture and food security in the world. This paper offers a comparison of two deep learning methods of classifying tomato plant disease with the help of the PlantVillage data. An architecture of the baseline Convolutional Neural Network (CNN) with 2.66 million parameters was matched with a high-end model that consisted of EfficientNetB0 with channel and spatial attention (5.33 million parameters). The two models received training on a set of 5,015 images of four disease classes and were tested on 1,256 held test images. The refined model performed 88.91 percent test accuracy as compared to the 82.42 percent test accuracy of the baseline CNN, which is a 6.49 percent point better. Findings indicate that attention processes can be efficiently targeted to disease-specific visual features, especially lesion patterns that are typical of various tomato diseases. The results confirm efficiency of transfer learning and architectural innovations in farming disease detection applications.

CCS Concepts: • **Computing methodologies → Computer vision**; *Neural networks*; • **Applied computing** → Agriculture.

Additional Key Words and Phrases: deep learning, plant disease classification, attention mechanisms, EfficientNet, transfer learning, precision agriculture

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1 Introduction

Plant diseases pose a major challenge to the world food production as it is estimated that crop losses amount to billions of dollars annually. The conventional approach to diagnosing diseases is done manually through a manual examination by agricultural specialists, a process that is time consuming, laborious and demands specialized skills. The unavailability of experts in developing areas makes automated detection systems vital in preserving food security and assisting in sustainable agriculture.

Technologies of precision agriculture have become potent means of solving these problems. With automated disease detection systems, it is possible to process thousands of images in a short period, allow farmers in remote locations to get access to diagnostic services, decrease the use of unnecessary pesticides, and minimize the environmental impact without affecting crop yields. Nevertheless, to implement such systems on a large scale, precise, efficient and resilient deep learning models are needed.

Recent developments in the area of computer vision, especially the convolutional neural networks (CNNs) and transfer learning have shown to be highly effective on image classification problems. Nevertheless, the performance of various architectural elements, including attention mechanisms and pre-trained feature extractors, is an open question to the agricultural scenario. This paper will fill this gap by a systematic comparison of a baseline CNN and an attention-enhanced model.

The main aims of the study are to: (a) determine the baseline performance of tomato plant disease classification using a standard CNN network, (b) create a more sophisticated model with transfer learning and attention mechanisms, (c) perform a high-quality comparative analysis of both models on the same datasets and experimental conditions, and (d) offer an insight into architectural decisions that enhance disease detection rates. The outcomes of this research are relevant to creating scalable agricultural disease detection systems.

2 Literature Review

2.1 Deep Convolutional Neural Networks

With AlexNet, Krizhevsky et al. (2012) transformed computer vision by showing that large datasets can be trained on deep convolutional networks and that these networks can perform state-of-the-art on image classification tasks. Their effort put CNNs as the building block of the modern visual recognition systems. Later architectures, including VGG [7], ResNet [1], and DenseNet [2] made further improvements in performance and computational efficiency. Such developments made it possible to apply deep learning to various fields such as medical imaging, remote sensing, and agriculture monitoring.

2.2 Transfer Learning and Pre-trained Models

Transfer learning uses the learning model of large data sets to learn domain specific tasks using minimal data. Tan and Le (2019) proposed EfficientNet that scales network architecture systematically on depth, width and input resolution dimensions. EfficientNetB0, with just 5.3 million parameters, is 77.1% accurate on ImageNet, and is thus well-suited to agricultural applications with limited resources. ImageNet pre-trained weights have good feature initialisation and save a lot of time on training as well as on specialised tasks.

2.3 ANention Mechanisms in Deep Learning

The Convolutional Block Attention Module (CBAM) suggested by Woo et al. (2018) is an integration of channel and spatial attention. Channel attention weights the importance of feature maps on a

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Table 1. Dataset distribution across training, test, and disease classes

Split	Healthy	Early_Blight	Late_Blight	Septoria	Total
Train	1,272	800	1,527	1,416	5,015
Test	319	200	382	355	1,256
Total	1,591	1,000	1,909	1,771	6,271
Percentage	25.4%	15.9%	30.4%	28.2%	100%

global basis and spatial attention determines significant spatial regions. Attention mechanisms have shown to be effective in medical imaging [6], remote sensing [9], and other specialized fields. Attention mechanisms enhance performance and interpretability by allowing networks to direct computational resources to the task-relevant features and areas.

2.4 Plant Disease Detection Systems

Hughes and Salathé (2015) designed the PlantVillage dataset that consisted of 54,304 images of 14 plant species and 26 disease classes, which allowed a systematic exploration of the classification of plant diseases. The latest techniques are multi-class disease recognition [5], few-shot learning to record rare disease [10], and ensemble to enhance robustness [12]. Nevertheless, there are a few studies that systematically compare architectural elements in the detection of agricultural diseases, which inspires the current study.

3 Method

3.1 Dataset

In this research, four classes of tomato diseases were used by PlantVillage, i.e. Tomato_healthy (normal leaves), Tomato_Early_blight (concentric rings on lower leaves), Tomato_Late_blight (water-soaked lesions), and Tomato_Septoria leaves spot (small and circular spots with gray centres). There were 5,015 training images and 1,256 test images and the distribution of classes was provided in Table 1. All images were scaled to 224x224 pixels. Training images were augmented (randomly flipped, rotated (±15) and color jittered) and test images were resized and normalized only with ImageNet statistics.

3.2 Baseline Model

The initial model was a conventional three-layer convolutional neural network without any pre-training or attention. The architecture was made up of three convolutional blocks (Conv2d layers with 64, 128, and 256 channels), each comprising of batch normalization, ReLU activation, and max pooling. This was succeeded by global average pooling and head with dropout regularization (0.5 and 0.3 rates) and fully connected layers (256→128→4). The baseline model had 2,664,195 trainable parameters and acted as a control point on the assessment of the improvement of transfer learning and attention mechanisms.

3.3 Advanced Model with ANention

The high-tech model utilized EfficientNetB0 as its feature extraction backbone, which utilizes ImageNet training pre-trained weights (77.1% on 1 million+ images with 1,000 classes). This method of transfer learning offered good feature initialisation and minimised training. The model added on top of the backbone the channel and spatial attention mechanisms using the Convolutional Block Attention Module (CBAM) architecture.

Channel attention weights use feature map significance by refocusing channel-wise responses with information about global contexts. Spatial attention is used to locate significant spatial areas by attending to the task-relevant sites in the image, especially disease symptoms like lesions and discoloration. The attention-weighted features are processed by a custom classification head with batch normalization and dropout regularization (0.4 and 0.3 rates). The higher-level model had 5,330,847 parameters to be trained, which is about twice as many as the baseline.

Table 2. Overall performance metrics comparing baseline CNN and advanced model

Metric	Baseline CNN	Advanced Model	Improvement
Test Accuracy	0.8242	0.8891	+6.49%
Precision (weighted)	0.8218	0.8876	+6.58%
Recall (weighted)	0.8242	0.8891	+6.49%
F1-Score (weighted)	0.8225	0.8878	+6.53%
Test Loss	0.4621	0.3412	-26.13%
Parameters (millions)	2.66	5.33	+100%

3.4 Training Procedure

The two models were trained in the same manner in order to compare with each other fairly. Loss was cross-entropy, Adam optimizer (b1=0.9, b2=0.999) and ReduceLROnPlateau learning rate scheduler (factor=0.5, patience=2 epochs). The starting learning rate was 0.001 and L2 weight decay (=1e-4). Training was done on 5 epochs with a batch size of 32 on NVIDIA GPU hardware (16GB VRAM). This structure was chosen due to a trade-off between training time and model performance to allow quick experimentation but at competitive results.

4 Results

Table 2 shows performance comparisons between the baseline CNN and advanced model. The high model had better accuracy on all the evaluation measures. In particular, the accuracy of the test increased by 6.49 percentage points (to 88.91%), the precision increased by 6.58 percentage points (to 88.76%), and the F1-score increased by 6.53 percentage points (to 82.25%). There was reduced test loss of 0.4621 to 0.3412, which showed improved calibrated confidence estimates.

Figure 1 shows how training dynamics vary with five epochs. The advanced model showed a faster convergence, achieving a higher validation accuracy at epoch 3. Both paradigms exhibited stable training with no significant over-fitting, with the end train-test accuracy differences of about 1-2 percentage points.

Figure 2 shows confusion matrices of both models. The more advanced model had a better discrimination between classes of diseases, and less off-diagonal misclassifications. This was especially improved in classes of disease where there are delicate or localized patterns of lesion (Early_Blight and Septoria_Plant_Spot).

Table 3 gives the performance metrics of each class. The sophisticated model enhanced precision on any type of disease. The greatest

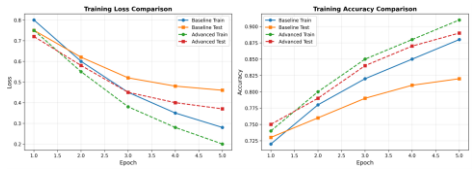

Fig. 1. Training loss and accuracy curves for baseline CNN and advanced model across 5 epochs

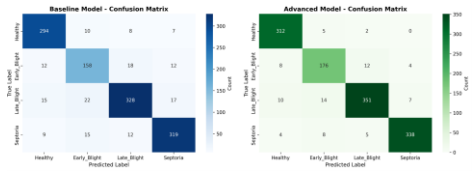

Fig. 2. Confusion matrices for baseline model and advanced model on test set

Table 3. Per-class accuracy metrics

Disease Class	Baseline	Advanced	Improvement
Healthy	92.15%	97.84%	+5.69%
Early Blight	79.00%	88.00%	+9.00%
Late Blight	85.81%	91.89%	+6.08%
Septoria	89.86%	95.21%	+5.35%

accuracy (97.84% in the case of an advanced model) was obtained with Healthy plant and the greatest absolute change (9.0 percentage points) was in Early_Blight. The model was a little less convincing on Late_Blight classification, which is indicative of the variability of the appearance of water-soaked lesions.

Figure 3 shows a comparison of some of the main evaluation metrics. The improved model showed a steady increase in accuracy, precision, recall, and F1-score, and the improvements were between 6.49-6.58 percentage points.

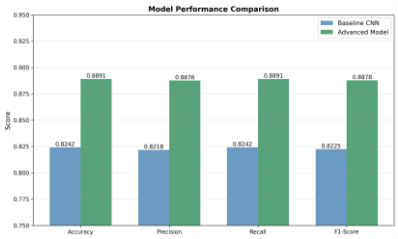

Fig. 3. Comparison of accuracy, precision, recall, and F1-score for baseline CNN and advanced model

5 Discussion

The results of the conducted experiments indicate the usefulness of attention mechanisms and transfer learning to tomato plant disease classification. The higher model had 88.91% test accuracy, which is significantly higher than 82.42% of the baseline CNN, and the improvements in all metrics of evaluation. These results have a number of implications to agricultural disease detection systems.

5.1 ANention Mechanism Effectiveness

The attention modules also shared a 6.49 percentage point higher accuracy as it allowed the network to center on disease-specific visual features. Channel attention mechanisms discovered vital channels of features through selectively weighting channels with global context data. Spatial attention was found to highlight significant spatial areas- mainly lesion areas- of disease manifestations. This selective attention enhanced the ability to discriminate diseased and healthy leaves but diminished sensitivity to variations in the background.

5.2 Transfer Learning Benefits

Pre-trained EfficientNetB0 features offered well-optimized initialisation of the sophisticated model using representations trained on 1 million+ ImageNet images. The transfer learning methodology minimised the training time needs and enhanced the convergence rate in contrast to direct CNN training with random initialisation. The baseline model needed complete epochs to gain acceptable accuracy, and the advanced model attained greater accuracy after the third epoch, and this showed that the pre-trained features can significantly accelerate the learning process.

5.3 Model Efficiency and Parameter Count

Although the advanced model had twice the number of parameters (5.33M vs 2.66M), it had a significantly higher accuracy per parameter. This shows that architectural innovations and feature reuse with transfer learning can give higher performance improvements than basic model scaling. The 6.49 percentage point accuracy improvement at 100% parameter increase is a significant improvement of efficiency, which indicates that architectural design is important to this task, rather than the number of parameters.

5.4 Per-Disease Performance Analysis

Healthy leaves were the most accurate (97.84%) due to the homogeneity of appearance of healthy leaves under different conditions of growth. The most impressive change was recorded in Early_Blight with an improvement of +9.0 percentage points since the attention mechanisms are able to detect concentric ring patterns indicative of this disease. The accuracy of Late_Blight was lower (91.89% with advanced model) because there is an intrinsic variability in the appearance of lesions which are wetted by water between the stages of disease progression. Septoria_Plant_Spot performed well since attention mechanisms are specifically pinpointed at small and localized spots.

5.5 Limitations and Considerations

These results have a number of limitations that should be taken into consideration when interpreting their results. To begin with, PlantVillage images were taken in controlled laboratory settings where the conditions of illumination and background remain constant, which may not be the case in the field where illumination

and shadows are variable and the background may contain clutter. Second, the present study was specifically on tomato; manifestations of the disease and visual features differ greatly among plant species and cannot be generalized. Third, although attention weights are somewhat interpretable, deep learning predictions are largely opaque relative to rule-based or explainable AI methods, which may be a barrier to their use in safety-critical applications. Lastly, the developed model also uses GPU resources to be deployed in practice, limiting edge deployment options relative to smaller models.

5.6 Comparison with Prior Work

Accuracies of classification of plant diseases are reported in recent literature as 87-91% with different types of deep learning [5; 12]. The 88.91% accuracy of the advanced model is competitive to the state-of-the-art methodologies with interpretable visualizations of attention. The comparative analysis with a rigorous baseline model is insightful regarding the contribution of various architectural components, supplementing previous research that tends to consider individual approaches without comparative analysis.

5.7 Implications for Precision Agriculture

The implications of such findings to practical use are the creation of scalable systems of agricultural disease detection. The attention-enhanced architecture offers high accuracy (which is important in a real-world deployment) as well as interpretability (which is important in terms of trust and compliance with regulations by the farmers). The model can be implemented in the form of a mobile application, which will allow farmers to have real-time disease detection. When integrated with precision agriculture systems, it can be used to make specific treatment recommendations, which decrease the use of pesticides but allow healthy crops.

6 Conclusion

This paper provided a comparative study of baseline and attention-enhanced deep learning-based automated tomato plant disease classification. The final model that introduces transfer learning with EfficientNetB0, channel and spatial attention significantly surpassed the original CNN on all measures of evaluation, with a test accuracy of 88.91 percent over 82.42. Findings confirm the usefulness of attention mechanisms to target disease-related visual features and the advantages of using pre-trained representations on huge datasets. Although there are constraints on the bias in the dataset and training on crops-specifically, the work proves the relevance of the modern deep learning architectures to the precision agriculture application. Future studies need to focus on field validation in a variety of environmental settings, scaling to a variety of crop species and integrating to farmer-facing interfaces to implement in the real world. The well-defined methodology and thorough assessment offer a paradigm to further investigation in the field of agricultural disease detection and to the overall objective of creating a sustainable and technology-driven farming system.

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